Supplementary Material for LABEL: Literature Asymmetry between Efficacy and Limits in Drug Knowledge and Language Models

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A Measuring Whether Text Names a Fact

Names. Each disease has three sets of names. The exact name is its MONDO label (Vasilevsky et al. 2022). Alternative names are the titles of Wikipedia redirects to the disease's article, which include synonyms, lay terms and abbreviations. Family names are the labels of the disease's MONDO ancestors up to two levels above it, taken from a directed child-to-parent table of 17,801 edges over 11,788 diseases. The sets are cumulative: a text that names the disease at a finer level is also counted at every coarser one, and each edge is assigned the finest level it reaches.

Matching rule. A phrase matches when it occurs on word boundaries, ignoring case, with an optional trailing "s" to allow plurals. Multi-word phrases match across any run of whitespace. Phrases shorter than four characters are never matched, so short abbreviations do not produce coincidental hits. Under this rule "asthma" does not match "asthmatic," and "colitis" does not match inside a longer word.

Floor. For every edge, three diseases that the drug has no edge to in that relation are drawn from the same relation's targets, and the same matcher is run on the drug's text for them. The share of these unrelated diseases that match is the floor. It estimates how often a disease name appears in a drug's text by coincidence rather than as a statement of the relation. Table 1 gives the exact-name floor on the probe sample.

Relation	Real edge	Floor
Indication	17.17	1.22
Off-label use	10.50	0.72
Contraindication	5.50	0.78

Table 1: Exact-name match rate (%) in Wikipedia for real edges and for three unrelated diseases per edge, 600 edges per relation.

Contribution of each level. On the same draw in Wikipedia, averaged over the three relations, adding alternative names to the exact name raises the match rate by 1.39 points. Adding the disease's MONDO parent raises it by a further 5.50 points, and adding ancestors up to two levels by 8.43 points. Each increase is the difference of floor-corrected rates, with the floor recomputed at every level, because wider name sets also match unrelated diseases more often. Only the exact and alternative names count as naming the fact in the main paper.

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B Corpora

Wikipedia. For each drug we fetch the current English Wikipedia article by title and cache its source text. The text measurement runs on the article as fetched.

PubMed. Abstracts are retrieved through the NCBI E-utilities (Sayers 2009) with the query `"<drug>"[Title/Abstract] AND (<fragment>)`, sorted by relevance. Each drug receives a budget of 40 abstracts, split evenly between two fragments, one framed for therapeutic use and one for adverse effects and contraindications, so that each relation receives 20. The union of returned records is fetched once and cached. A request that fails is not cached, so a failure is never recorded as an empty result. The first version of the corpus used a single relevance-ranked query of 40 abstracts per drug; the comparison of the two versions is reported in the main paper.

C Model Probe

Question format. Each edge is scored as a completion. The prompt is a relation-specific stem that names the drug and stops before the disease (Table 3); each option is appended as a continuation, and its score is the mean token log-probability of the continuation. The option with the highest score is the model's answer. No chat template, passage or retrieval is used.

Options. Each question has five options: the true disease and four distractors drawn at random from the same relation's targets in the sample. Distractors exclude every disease the drug is paired with in that relation within the sample. Partners outside the sample cannot be excluded; with about 600 targets per relation and a handful of partners per drug, the chance that a distractor is an unrecorded true partner is on the order of one percent, and it applies equally to all relations. The order of options is shuffled with a fixed seed.

Relation	Stem
Indication	*X is a drug used to treat the disease called*
Contraind.	*X is a drug that must not be given to patients who have*
Off-label use	*X is a drug sometimes used off-label to treat*
Treats (Hetionet)	*X is a drug used to treat the disease called*
Palliates (Hetionet)	*X is a drug used to relieve the symptoms of*

Table 2: Prompt stems; X is the drug's name.

Models. Qwen2.5-0.5B-Instruct and Qwen2.5-7B-Instruct (Qwen Team 2024) run in 32-bit floating point on CPU with Hugging Face Transformers.

D Prompt Stems and Declines

Relation	Prompt stem (X is the drug's name)
Indication	*X is a drug used to treat the disease called*
Contraindication	*X is a drug that must not be given to patients who have*
Off-label use	*X is a drug sometimes used off-label to treat*
Treats	*X is a drug used to treat the disease called*
Palliates	*X is a drug used to relieve the symptoms of*

Table 3: Prompt stems for the closed-book probe. Each option is scored as the continuation of the stem; the last two rows are for the Hetionet relations.

	Indication		Contraindication	
Page	closed	page	closed	page
Names the disease	6.0	5.8	36.2	41.6
Names only its family	9.1	26.0	42.1	54.4
Silent	29.6	55.6	43.3	76.2

Table 4: Share of questions answered "Not enough information" (%), closed book and with the drug's page, averaged over Qwen2.5 7B, OLMo 2 7B, and Granite 3.1 8B.

E Negative Control on Hetionet

F Harder-Distractor Check

Questions. The check reuses the question set of the companion abstention study. Each question asks "Which disease is X used to treat?" (indication) or "In patients with which disease must X not be given?" (contraindication) and lists five diseases and the option "Not enough information to tell," lettered A to F. The position of each option, including the decline option, is random.

	treats	*palliates*
Named in Wikipedia (%)	57.33 (n=600)	56.10 (n=385)
Odds ratio, pooled	1.05, $p = 0.70$	
Odds ratio, within drug (51)	2.48 [1.29, 4.75]	
Relation term, degree held fixed	$p = 0.16$	
Gap by edges per drug: 1–2	$+8.16$ ($p = 0.06$)	
3–5	$+0.93$ ($p = 0.89$)	
6–12	-11.38 ($p = 0.32$)	

Table 5: Negative control on Hetionet. Two benefit relations, same corpus and matcher. No gap survives the degree control, and the strata change sign.

Distractors. Wrong options are taken first from the true disease's MONDO siblings, then from its cousins (other children of its grandparent), and only then from random diseases of the same relation. They never include the true disease's own ancestors or descendants, or any disease the drug is related to by the same relation. In 481 of 856 questions all four distractors come from siblings or cousins.

Scoring. Each prompt is formatted with the model's chat template, and the answer is the option letter with the highest next-token probability. Both spellings of each letter, with and without a leading space, are scored and the better one kept; probabilities are renormalized over the six letters. Prompts are batched with left padding, and position indices are rebuilt from the attention mask so that batched and single scores agree.

Models. Qwen2.5 0.5B and 7B, OLMo 2 7B (Team OLMo et al. 2024) and Granite 3.1 8B (IBM Granite Team 2024), all instruction-tuned, in 32-bit floating point on CPU.

G Statistical Procedures

Within-drug odds ratio. For drug d, let a_d and b_d be its named and unnamed indication edges, c_d and e_d its named and unnamed contraindication edges, and n_d their total. The Mantel–Haenszel estimate (Mantel and Haenszel 1959) is

$$\mathrm{OR}_{\mathrm{MH}} = \frac{\sum_d a_d e_d / n_d}{\sum_d b_d c_d / n_d}, \tag{1}$$

with a 95% confidence interval $\exp(\log \mathrm{OR}_{\mathrm{MH}} \pm 1.96/\sqrt{\sum_d V_d})$, where $V_d = n_{d,1} n_{d,2} m_{d,1} m_{d,2} / (n_d^2 (n_d - 1))$ is the hypergeometric variance of drug d's table with row totals $n_{d,\cdot}$ and column totals $m_{d,\cdot}$, and a continuity-corrected Cochran–Mantel–Haenszel test of no association.

Edge-count control. A logistic model regresses whether a fact is named on an indicator for the relation and on $\log$ of the drug's number of edges in that relation in the full graph. The relation coefficient tests the gap with edge count held fixed.

Wilson intervals. For k successes in n trials, $\hat{p} = k/n$, and $z = 1.96$, the interval is

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}. \tag{2}$$

Paired gain from scale. For each edge i, the gain is $g_i = y_i^{7B} - y_i^{0.5B}$, where $y = 1$ for a correct answer. The statistic is the mean gain on indications minus the mean gain on contraindications. Its interval comes from 5,000 bootstrap resamples in which drugs, not edges, are drawn with replacement, so that edges of the same drug move together. The p-value uses the normal approximation with the bootstrap standard error.

Gap within naming groups. In the distractor check, the indication-minus-contraindication difference is computed separately within questions whose text names the answer, names only its family, or names nothing. The group differences δ_g are combined as $\sum_g w_g \delta_g / \sum_g w_g$ with $w_g = n_{g,1} n_{g,2} / (n_{g,1} + n_{g,2})$, and the standard error follows from the binomial variances of the group proportions.

H Additional Results

Relation	0.5B [95% CI]	7B [95% CI]	Gain
Indication	31.50 [27.91, 35.32]	50.83 [46.84, 54.82]	+19.33
Off-label	28.50 [25.03, 32.24]	35.00 [31.29, 38.90]	+6.50
Contraind.	22.17 [19.03, 25.66]	23.83 [20.60, 27.40]	+1.67

Table 6: Probe accuracy (%) and paired gain from 0.5B to 7B (points), 600 edges per relation. Indication gain minus contraindication gain: +17.67 [12.68, 22.70], $p = 4.8 \times 10^{-12}$.

Edges per drug	3–5	6–12	13+
Wikipedia	+13.94	+6.04	+1.64
Wikipedia + PubMed	+11.81	+6.90	+2.23

Table 7: Indication minus contraindication coverage (points) within strata of the drug's number of edges in the relation. The 13+ stratum is not distinguishable from zero in either corpus.

Model	Ind.	Contra.	Gap	Within	p (within)
Qwen2.5 0.5B	36.0	28.1	+7.9	+8.2	8.9×10^{-3}
Qwen2.5 7B	56.0	25.1	+30.9	+31.3	$< 10^{-15}$
OLMo 2 7B	56.0	33.0	+23.0	+23.6	2.3×10^{-13}
Granite 3.1 8B	58.7	12.8	+45.9	+46.3	$< 10^{-15}$

Table 8: Harder-distractor check, closed book (%). 450 indication and 406 contraindication questions. Raw-gap p-values are 0.013 for Qwen2.5 0.5B, 1.5×10^{-11} for OLMo 2 7B, and below 10^{-15} for Qwen2.5 7B and Granite 3.1 8B.

Control graph. On Hetionet, Wikipedia names the disease for 57.33% of 600 sampled *treats* edges and 56.10% of 385 measurable *palliates* edges, an odds ratio of 1.05 ($p = 0.70$). Within the 51 drugs that carry both relations, the Mantel–Haenszel odds ratio is 2.48 (95% CI 1.29–4.75). With the drug's edge count held fixed, the relation term is not significant ($p = 0.16$), and the edge-count strata show no consistent direction (+8.2 points for 1–2 edges, +0.9 for 3–5, -11.4 for 6–12). In PrimeKG, the relation term remains significant under the same control ($p = 2.4 \times 10^{-4}$ in Wikipedia). Hetionet uses the Disease Ontology rather than MONDO for its hierarchy.

I Reproducibility

Pipeline. The analysis runs in five stages, each a standalone script that reads the previous stage's files: building the edge samples from PrimeKG, Hetionet and MONDO; fetching and caching Wikipedia articles and PubMed abstracts; measuring the naming level of every edge against its floor; scoring the model probe; and computing the within-drug, paired and distractor statistics. A single script runs all five stages from the public downloads, submits the model jobs to the cluster scheduler, and writes every table. The statistics use the Python standard library only. Random draws use a fixed seed of 0.

Compute. All model scoring runs on CPU nodes of a university supercomputer, with PyTorch 2.14 and Transformers 5.20 in 32-bit floating point. No GPU is required.

Data. PrimeKG, Hetionet, MONDO, Wikipedia and PubMed are public. We release the sampled edge lists, the per-edge naming levels and the per-edge model scores, together with the scripts, so that every table in the paper and this supplement can be regenerated from the released files.

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